



**UNIVERSITY OF
CALGARY**

ENGG 684- Introduction to Project Management

Segment 3: Cost Estimation and Cash Flow

Shaktipal Singh Deora (UCID 30239938)

Group – 8, M&M Constructions Ltd.

Course Instructor:

Estacio Pereira, PhD

Assistant Professor (Teaching) in Project Management

Department of Civil Engineering – Schulich School of Engineering

1. Direct and Indirect costs Calculation

Based upon the data provided in document “Course Project” the following table has been calculated for Direct cost with Task based per Project for the contractors which I selected in segment 2. I have also bifurcated the Direct cost, Indirect cost (In Cash Flow Analysis) for each week for the selected contractors and the same will be attached in upcoming sections.

Table 1: Direct Cost Calculation table with Task based selected sub-contractor cost per Project

Task	Subcontractor	Weeks Rounded for Project 1	Weeks Rounded for Project 2	Weeks Rounded for Project 3	Weeks Rounded for Project 4	Weeks Rounded for Project 5	Weeks Rounded for Project 6	Weeks Rounded for Project 7	Weeks Rounded for Project 8	
Architectural Drawing	Arch ContB	2	3	4	3	3	4	2	5	
Geotechnical	Geo ContB	2	3	4	2	2	3	4	4	
Structural Drawing	Struct ContA	1	3	3	2	2	2	2	3	
Electrical	Elec ContA	1	3	3	2	2	3	1	3	
Mechanical	Mech ContA	2	3	4	3	3	4	2	4	
Plumbing	Plumb ContA	2	3	4	2	3	3	2	4	
Compatibility Check	Comp ContB	2	3	4	2	2	3	2	4	
Task	Suncontractor	Direct cost for Project 1	Direct cost for Project 2	Direct cost for Project 3	Direct cost for Project 4	Direct cost for Project 5	Direct cost for Project 6	Direct cost for Project 7	Direct cost for Project 8	Total cost for Each Task for all the projects
Architectural Drawing	Arch ContB	20,000	30,000	40,000	30,000	30,000	40,000	20,000	50,000	260,000
Geotechnical	Geo ContB	18,000	27,000	36,000	18,000	18,000	27,000	36,000	36,000	216,000
Structural Drawing	Struct ContA	9,000	27,000	27,000	18,000	18,000	18,000	18,000	27,000	162,000
Electrical	Elec ContA	10,000	30,000	30,000	20,000	20,000	30,000	10,000	30,000	180,000
Mechanical	Mech ContA	16,000	24,000	32,000	24,000	24,000	32,000	16,000	32,000	200,000
Plumbing	Plumb ContA	14,000	21,000	28,000	14,000	21,000	21,000	14,000	28,000	161,000
Compatibility Check	Comp ContB	16,000	24,000	32,000	16,000	16,000	24,000	16,000	32,000	176,000
Task-based selected subcontractor costs per project		103,000	183,000	225,000	140,000	147,000	192,000	130,000	235,000	1,355,000

And similarly, I also calculated Indirect cost for each week are tabulated below

Table 2: Indirect Cost Calculation Table

Category	Description	Quantity	Unit Cost per Week (\$)	Subtotal Week
Office Rent	Lease, utilities, shared workspace	2 rooms	5000	10000
Software Licenses	BIM, CAD, scheduling, simulation tools	8 licenses	100	800
Administrative Staff Salaries	Project coordinator, HR, finance	2 staff	800	1600
Office Supplies & Printing	Stationery, printing, largeformat plotting	Flat weekly cost	500	500
IT Infrastructure & Support	Cloud storage, hardware, technical support	3 systems	60	180
Communication Tools	PM software, video conferencing tools	3 licenses	30	90
Professional Liability Insurance	Design liability coverage	1 company policy	135	135
Training & Professional Development	Weekly average for webinars, training, certifications	Flat weekly cost	70	70
Miscellaneous Overhead	Travel, subscriptions, small equipment, snacks	Flat weekly allowance	1000	1000
Total Cost per week				14375

And the contingency cost which I calculated using the Monte Carlo simulation is \$198,000 (and the calculation for which will be explained in Contingency Calculation – Monte Carlo Simulation), and the total direct cost for each selected contractor for all the projects are \$1,355,000 and the Indirect cost for total weeks for each project to complete (36 weeks) is \$517,500 (\$14,375 x 36) and the total sum of which will be \$2,070,500.

2. Contingency Calculation – Monte Carlo Simulation

Contingency plays an important role in project management as it provides a safeguard against uncertainty. It typically refers to an allocated buffer of money, time, or resources that can be used to address unexpected events or risks that may arise during the project. In our case, we are focusing on cost contingency rather than time. The objective of contingency is to ensure the project can still meet its goals and remain on track even when unforeseen challenges occur.

To determine the contingency value for this project, I utilized the Excel tool provided by the professor titled **“Quantitative Risk Assessment.”** In this sheet, I identified the relevant risks and extracted their associated probabilities from the **Course Project V6** document. I then applied a **Monte Carlo simulation** to generate randomized probability outcomes based on the input risk data. The simulation results produced a range of possible cost impacts.

I ran the simulation for **2,556 iterations**, allowing the model to converge towards a reliable estimate of the expected cost exposure. From the generated probability distribution, I selected a **confidence level of 85%**, meaning that there is an 85% likelihood that the project cost will not exceed the estimated value including contingency. I will explain the rationale behind selecting this confidence level in a later section.

Based on this analysis, the calculated contingency amount is **approximately \$198,000**. A screenshot of the simulation results is also attached for clearer understanding.

Risk Number	Risk Description	Value	Probability	Random Probability	Final
1	Safety Incident (e.g., during site visit)	\$ 300,000.00	0.050	0.702	\$ -
2	Conflict Between Project Parties	\$ 90,000.00	0.150	0.440	\$ -
3	Inflation (software/tools)	\$ 15,000.00	0.450	0.437	\$ 15,000.00
4	Quality Issues – Redesign	\$ 90,000.00	0.100	0.177	\$ -
5	Design Scope Creep	\$ 45,000.00	0.300	0.658	\$ -
6	Delay in Geotechnical Report	\$ 36,000.00	0.200	0.610	\$ -
7	Software Licensing Delays	\$ 24,000.00	0.100	0.306	\$ -
8	Regulatory/Code Change	\$ 60,000.00	0.100	0.600	\$ -
9	Rework	\$ 54,000.00	0.150	0.565	\$ -
10	Design Software Crash/Data Loss	\$ 27,000.00	0.080	0.590	\$ -
11	Coordination Delay Between Disciplines	\$ 39,000.00	0.180	0.154	\$ 39,000.00
12	Unclear Client Requirements	\$ 42,000.00	0.220	0.082	\$ 42,000.00
13	Misinterpretation of Regulations	\$ 36,000.00	0.090	0.599	\$ -
14	Third-Party Consultant Delay (e.g., HVAC, acoustics)	\$ 37,500.00	0.140	0.409	\$ -
15	Incorrect Existing Conditions Assumptions	\$ 40,500.00	0.130	0.523	\$ -
16	Inadequate Site Survey or Data Error	\$ 45,000.00	0.110	0.442	\$ -
17	Team Turnover (new team member ramp-up)	\$ 30,000.00	0.160	0.447	\$ -
					\$ 96,000.00

Figure 1: Screenshot of the Random Probability to calculate contingency (Monte Carlo Simulation)

Based on the results generated from the simulation iterations, I derived five key contingency cost values corresponding to different confidence levels: the Average value, as well as the 25th percentile, 85th percentile, and 95th percentile, along with the Minimum estimated value. These values provide a clearer understanding of how the contingency amount varies depending on the level of risk tolerance we choose to adopt. A summary table containing these contingency values is provided below for reference.

Table 3: Contingency Cost for different confidence level

SRL.	Confidence Level	Contingency Cost
1	Average	\$ 124,818.08
2	95%	\$ 336,000.00
3	25%	\$ 60,000.00
4	85%	\$ 198,000.00
5	Min	\$ 0.00

I selected the 85% confidence level because it provides a balanced approach between risk tolerance and budget efficiency. Choosing a confidence level that is too low (e.g., 25%) would underestimate the potential cost impact and increase the likelihood of the project running over budget. On the other hand, selecting a very high confidence level (e.g., 95%) would significantly increase the contingency reserve, which may result in allocating more funds than necessary and reducing the project's cost efficiency.

An 85% confidence level is commonly accepted in project planning for projects of moderate complexity and uncertainty, such as this one because it reflects a proactive but realistic risk posture. It ensures that we are adequately prepared for reasonably probable unexpected events without excessively inflating the project budget. Therefore, 85% serves as a practical midpoint, balancing financial responsibility with risk mitigation, the overall contingency calculation based upon the number of iterations the Histogram is attached below for providing better understanding how the contingency cost is spread.

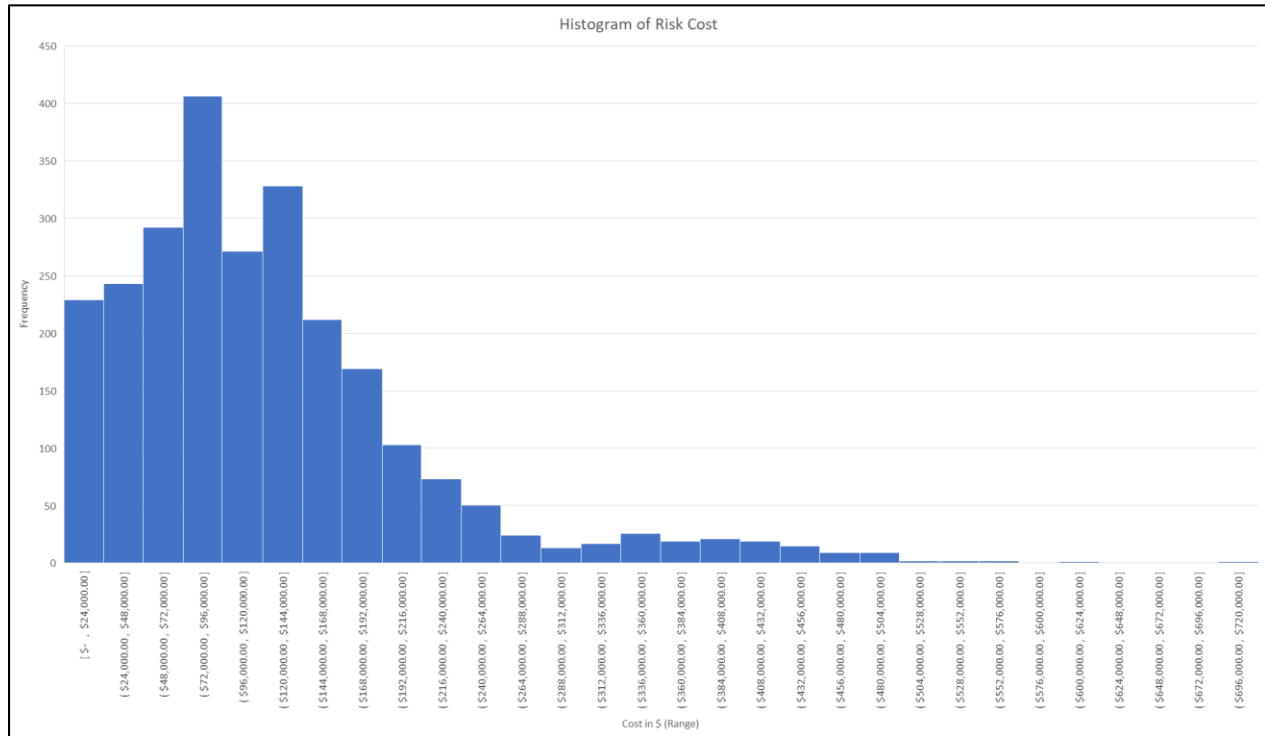


Figure 2: Histogram of Risk Cost

3. Cash flow Analysis

After determining the contingency amount, the next step was to perform the cash flow analysis using the Excel template provided by the professor. Before beginning the cash flow calculations, it was necessary to establish a few assumptions and considerations:

- Risk Allocation (Contingency Allocation): The contingency cost at the 85% confidence level was distributed evenly across the project duration of 36 weeks, which resulted in approximately \$5,500 per month.
- Mobilization Payment: A mobilization cost of \$500,000 was included at the start of the project.
- Interest Rate: An interest rate of 1.5% was assumed for financing calculations.
- Insurance and Risk-Related Costs: A combined insurance and risk allowance of \$19,000 was included to account for uncertainties such as inflation, software failure, and coordination delays between disciplines.
- Markup: A markup of 25% was applied, which is a common value for EPC (Engineering, Procurement, and Construction) projects.
- Retainage: A retainage rate of 100% was assumed, meaning the owner will release payment only upon completion of the project.
- Maintaining the total revenue fixed at \$3,200,000

This set of assumptions defines what I refer to as the "Original Scenario." In the following sections, I will also compare this with two alternative variation scenarios. The overdraft graph for the Original Scenario and the corresponding cash flow analysis table is attached below for reference.

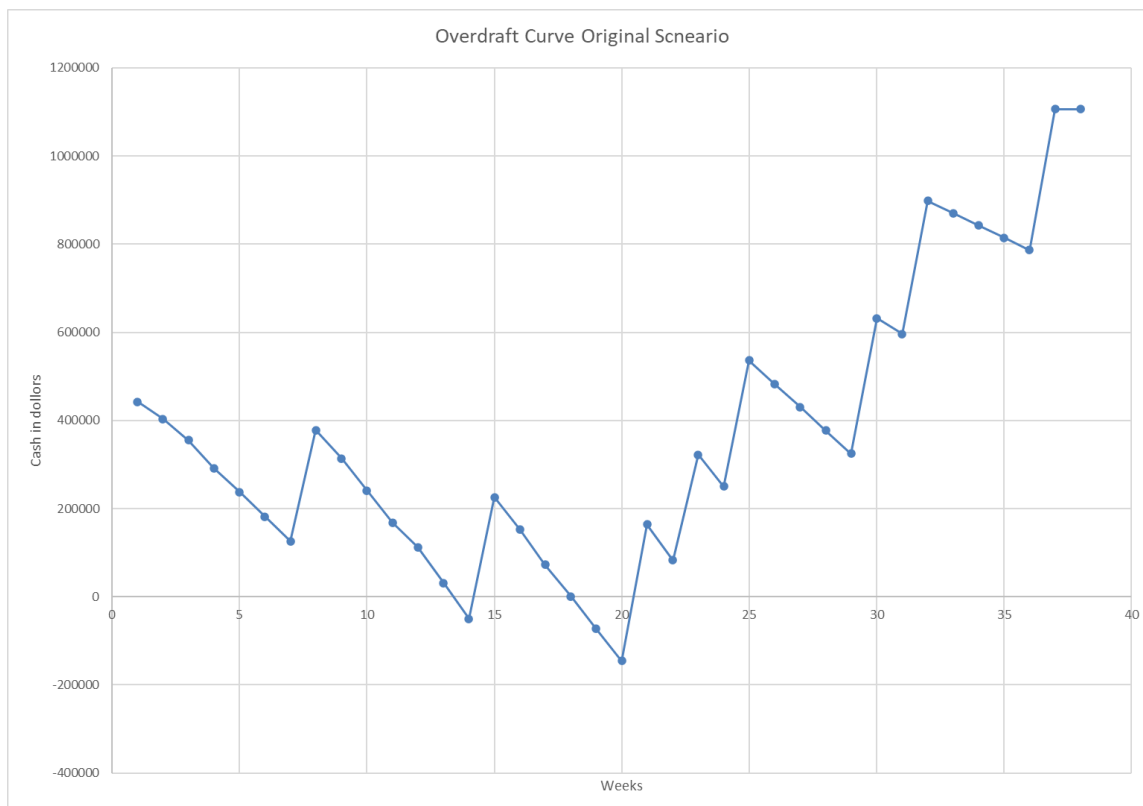


Figure 3: Overdraft Curve for Original Scenario

Table 4: Cash flow Analysis for Original Scenario

Original Scenario	Week	Direct Cost	Indirect Cost	Insurance	Risk Allocation	Subtotal - not included risk - not paid by the owner in this scenario	Markup (25%)	Total Billed	Retainage Applied (100%)	Payment Received	Total Cost to Date - Including Risk	Total Billed to Date	Total Paid to Date	Overdraft End of Month	Interest (1.5%)	Max Overdraft (to date)
Expenditure	1	19000	14375	19000	5500	52375	13093.75	65468.75	65468.75	500000	57875	65468.75	500000	-442125	0	-442125
Receive																
Expenditure	2	19000	14375	0	5500	33375	8343.75	41718.75	41718.75		96750	107187.5		-403250	0	-403250
Receive										0			500000			
Expenditure	3	28000	14375	0	5500	42375	10593.75	52968.75	52968.75		144625	160156.25		-355375	0	-355375
Receive										0			500000			
Expenditure	4	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		208500	233125		-291500	0	-291500
Receive										0			500000			
Expenditure	5	34000	14375	0	5500	48375	12093.75	60468.75	60468.75		262375	293593.75		-237625	0	-237625
Receive										0			500000			
Expenditure	6	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		318250	356562.5		-181750	0	-181750
Receive										0			500000			
Expenditure	7	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		374125	419531.25		-125875	0	-125875
Receive										300000			800000			
Expenditure	8	28000	14375	0	5500	42375	10593.75	52968.75	52968.75		422000	472500		-378000	0	-378000
Receive										0			800000			
Expenditure	9	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		485875	545468.75		-314125	0	-314125
Receive										0			800000			
Expenditure	10	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		558750	629687.5		-241250	0	-241250
Receive										0			800000			
Expenditure	11	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		631625	713906.25		-168375	0	-168375
Receive										0			800000			
Expenditure	12	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		687500	776875		-112500	0	-112500
Receive										0			800000			
Expenditure	13	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		768375	871093.75		-31625	0	-31625
Receive										0			800000			
Expenditure	14	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		849250	965312.5		49250	738.75	49988.75
Receive										340000			1140000			
Expenditure	15	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		913125	1038281.25		-226136.25	0	-226136.25
Receive										0			1140000			
Expenditure	16	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		986000	1122500		-153261.25	0	-153261.25
Receive										0			1140000			
Expenditure	17	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1066875	1216718.75		-72386.25	0	-72386.25
Receive										0			1140000			
Expenditure	18	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1138750	1299687.5		-511.25	0	-511.25
Receive										0			1140000			
Expenditure	19	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1210625	1382656.25		71363.75	1070.45625	72434.20625
Receive										0			1140000			
Expenditure	20	51000	14375	0	5500	65375	16343.75	81718.75	81718.75		1281500	1464375		143309.2063	2149.638094	145458.8443
Receive										380000			1520000			
Expenditure	21	51000	14375	0	5500	65375	16343.75	81718.75	81718.75		1352375	1546093.75		-163666.156	0	-163666.156
Receive										0			1520000			
Expenditure	22	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1433250	1640312.5		-82791.1557	0	-82791.1557
Receive										320000			1840000			
Expenditure	23	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1514125	1734531.25		-321916.156	0	-321916.156
Receive										0			1840000			
Expenditure	24	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1586000	1817500		-250041.156	0	-250041.156
Receive										350000			2190000			
Expenditure	25	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		1649875	1890468.75		-536166.156	0	-536166.156
Receive										0			2190000			
Expenditure	26	34000	14375	0	5500	48375	12093.75	60468.75	60468.75		1703750	1950937.5		-482291.156	0	-482291.156
Receive										0			2190000			
Expenditure	27	32000	14375	0	5500	46375	11593.75	57968.75	57968.75		1755625	2008906.25		-430416.156	0	-430416.156
Receive										0			2190000			
Expenditure	28	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1808500	2068125		-377541.156	0	-377541.156
Receive										0			2190000			
Expenditure	29	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1861375	2127343.75		-324666.156	0	-324666.156
Receive										360000			2550000			
Expenditure	30	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1914250	2186562.5		-631791.156	0	-631791.156
Receive										0			2550000			
Expenditure	31	16000	14375	0	5500	30375	7593.75	37968.75	37968.75		1950125	2224531.25		-595916.156	0	-595916.156
Receive										330000			2880000			
Expenditure	32	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		1978000	2252500		-898041.156	0	-898041.156
Receive										0			2880000			
Expenditure	33	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2005875	2280468.75		-870166.156	0	-870166.156
Receive										0			2880000			
Expenditure	34	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2033750	2308437.5		-842291.156	0	-842291.156
Receive										0			2880000			
Expenditure	35	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2061625	2336406.25		-814416.156	0	-814416.156
Receive										0			2880000			
Expenditure	36	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2089500	2364375		-786541.156	0	-786541.156
Receive										320000			3200000			
Expenditure	37	0	0	0	0	0	0	0	0		2089500	2364375		-1106541.16	0	-1106541.16
Receive										0			3200000			

4. Financing Scenarios and Payment Terms Sensitivity Analysis.

After completing the cash flow analysis for the Original Scenario, I conducted a sensitivity analysis by evaluating two variation scenarios to observe how changes in key financial conditions impact the overdraft and overall cash flow profile.

- Variation Scenario 1 – Payment Variation (Mobilization Payment):
In this scenario, I assumed that no mobilization payment was received at the start of the project. This allowed me to analyze how the absence of upfront funding influences the project's overdraft requirements and cash flow trends.
- Variation Scenario 2 – Financial Variation (Interest Rates):
In this scenario, I retained the mobilization payment but increased the interest rate from 1.5% to 20%. This was done to examine the impact of a significant rise in the cost of financing on the project's cash flow and overdraft behavior.

For each scenario, I generated the corresponding overdraft curves and cash flow analysis tables, which are attached below. In the next section, I will discuss the observed differences and explain how these variations influence project financing and risk exposure based on the attached results.

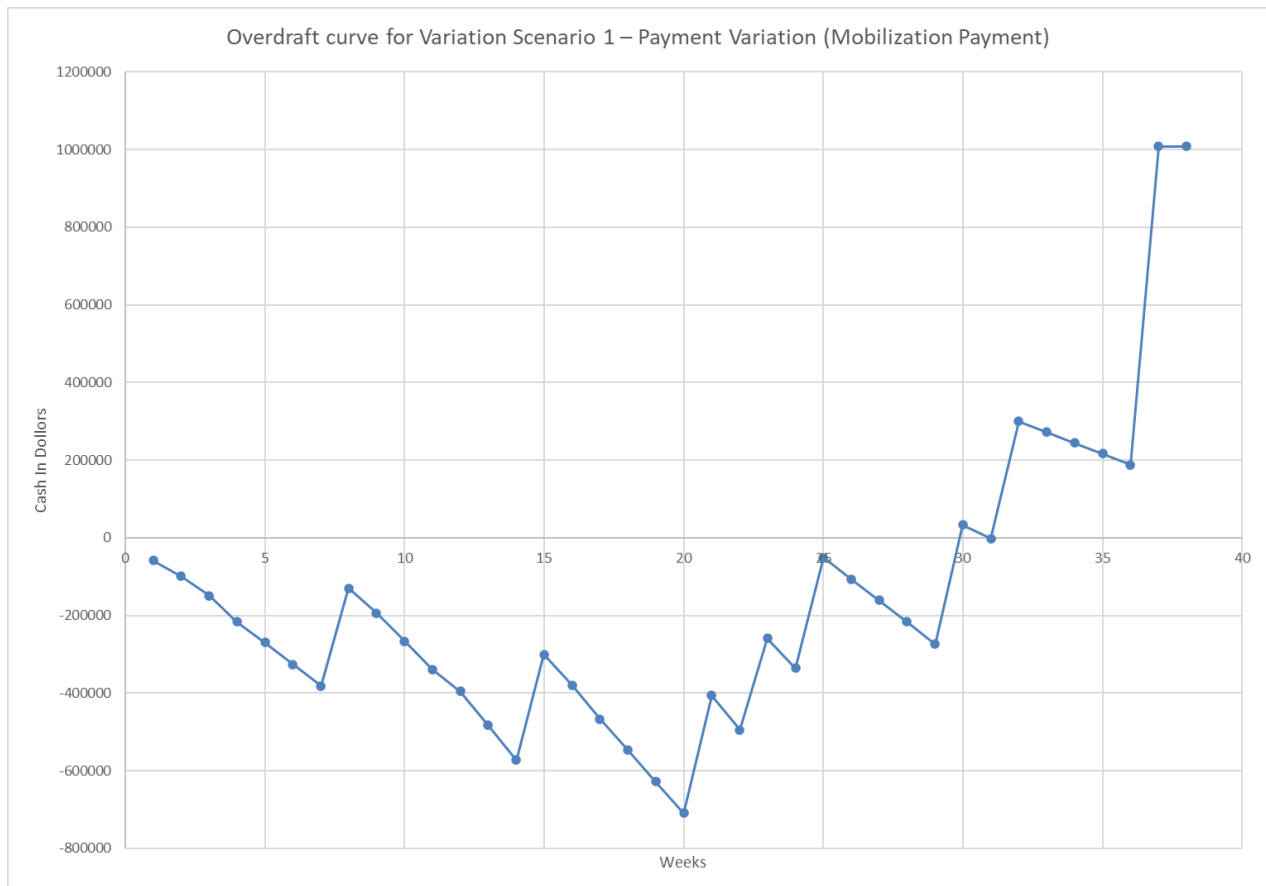


Figure 4: : Overdraft curve for Variation Scenario 1 – Payment Variation (Mobilization Payment)

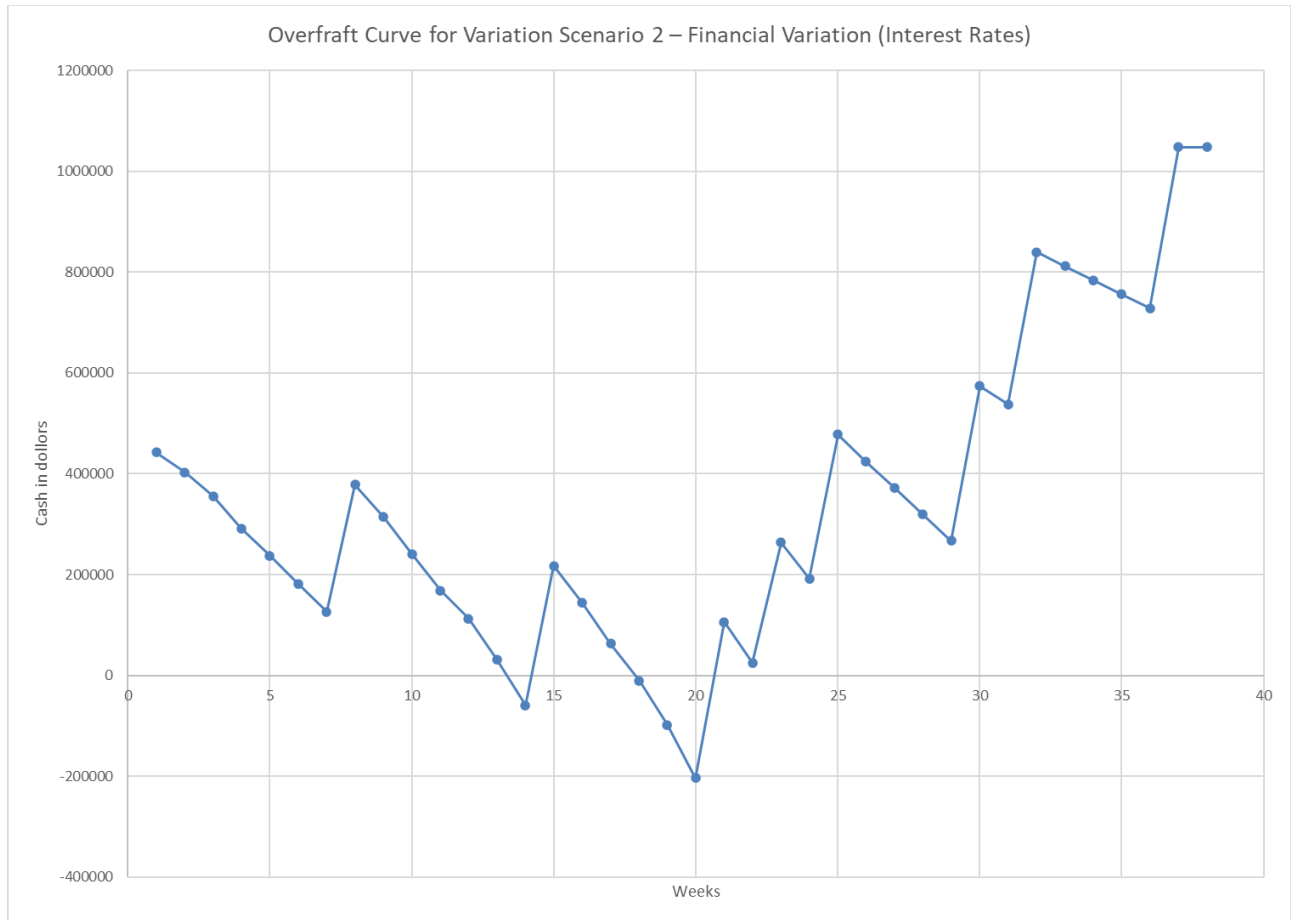


Figure 5: Overdraft Curve for Variation Scenario 2 – Financial Variation (Interest Rates)

Table 5: Cash Flow Analysis for Variation Scenario 1 – Payment Variation (Mobilization Payment)

Variation Scenario I	Week	Direct Cost	Indirect Cost	Insurance	Risk Allocation	Subtotal - not included risk - not paid by the owner in this scenario	Markup (25%)	Total Billed	Retainage Applied (100%)	Payment Received	Total Cost to Date - Including Risk	Total Billed to Date	Total Paid to Date	Overdraft End of Month	Interest (1.5%)	Max Overdraft (to date)
Expenditure	1	19000	14375	19000	5500	52375	13093.75	65468.75	65468.75	0	57875	65468.75	0	57875	868.125	58743.125
Receive																
Expenditure	2	19000	14375	0	5500	33375	8343.75	41718.75	41718.75		96750	107187.5		97618.125	1464.271875	99082.39688
Receive										0			0			
Expenditure	3	28000	14375	0	5500	42375	10593.75	52968.75	52968.75		144625	160156.25		146957.3969	2204.360953	149161.7578
Receive										0			0			
Expenditure	4	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		208500	233125		213036.7578	3195.551367	216232.3092
Receive										0			0			
Expenditure	5	34000	14375	0	5500	48375	12093.75	60468.75	60468.75		262375	293593.75		270107.3092	4051.609638	270107.3092
Receive										0			0			
Expenditure	6	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		318250	356562.5		325982.3092	4889.734638	325982.3092
Receive										0			0			
Expenditure	7	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		374125	419531.25		381857.3092	5727.859638	381857.3092
Receive										300000			300000			
Expenditure	8	28000	14375	0	5500	42375	10593.75	52968.75	52968.75		422000	472500		129732.3092	1945.984638	129732.3092
Receive										0			300000			
Expenditure	9	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		485875	545468.75		193607.3092	2904.109638	193607.3092
Receive										0			300000			
Expenditure	10	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		558750	629687.5		266482.3092	3997.234638	266482.3092
Receive										0			300000			
Expenditure	11	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		631625	713906.25		339357.3092	5090.359638	339357.3092
Receive										0			300000			
Expenditure	12	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		687500	776875		395232.3092	5928.484638	395232.3092
Receive										0			300000			
Expenditure	13	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		768375	871093.75		476107.3092	7141.609638	483248.9188
Receive										0			300000			
Expenditure	14	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		849250	965312.5		564123.9188	8461.858783	572585.7776
Receive										340000			640000			
Expenditure	15	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		913125	1038281.25		296460.7776	4446.911664	300907.6893
Receive										0			640000			
Expenditure	16	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		986000	1122500		373782.6893	5606.740339	379389.4296
Receive										0			640000			
Expenditure	17	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1066875	1216718.75		460264.4296	6903.966444	467168.3961
Receive										0			640000			
Expenditure	18	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1138750	1299687.5		539043.3961	8085.650941	547129.047
Receive										0			640000			
Expenditure	19	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1210625	1382656.25		619004.047	9285.060705	628289.1077
Receive										0			640000			
Expenditure	20	51000	14375	0	5500	65375	16343.75	81718.75	81718.75		1281500	1464375		699164.1077	10487.46162	709651.5693
Receive										380000			1020000			
Expenditure	21	51000	14375	0	5500	65375	16343.75	81718.75	81718.75		1352375	1546093.75		400526.5693	6007.89854	406534.4679
Receive										0			1020000			
Expenditure	22	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1433250	1640312.5		487409.4679	7311.142018	494720.6099
Receive										320000			1340000			
Expenditure	23	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1514125	1734531.25		255595.6099	3833.934148	259429.544
Receive										0			1340000			
Expenditure	24	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1586000	1817500		331304.544	4969.56816	336274.1122
Receive										350000			1690000			
Expenditure	25	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		1649875	1890468.75		50149.11219	752.2366829	50901.34887
Receive										0			1690000			
Expenditure	26	34000	14375	0	5500	48375	12093.75	60468.75	60468.75		1703750	1950937.5		104776.3489	1571.645233	106347.9941
Receive										0			1690000			
Expenditure	27	32000	14375	0	5500	46375	11593.75	57968.75	57968.75		1755625	2008906.25		158222.9941	2373.344912	160596.339
Receive										0			1690000			
Expenditure	28	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1808500	2068125		213471.339	3202.070085	216673.4091
Receive										0			1690000			
Expenditure	29	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1861375	2127343.75		269548.4091	4043.226137	273591.6352
Receive										360000			2050000			
Expenditure	30	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1914250	2186562.5		-33533.36476		-33533.36476
Receive										0			2050000			
Expenditure	31	16000	14375	0	5500	30375	7593.75	37968.75	37968.75		1950125	2224531.25		2341.635241	35.12452862	2376.75977
Receive										330000			2380000			
Expenditure	32	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		1978000	2252500		-299748.2402		-299748.2402
Receive										0			2380000			
Expenditure	33	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2005875	2280468.75		-271873.2402		-271873.2402
Receive										0			2380000			
Expenditure	34	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2033750	2308437.5		-243998.2402		-243998.2402
Receive										0			2380000			
Expenditure	35	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2061625	2336406.25		-216123.2402		-216123.2402
Receive										0			2380000			
Expenditure	36	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2089500	2364375		-188248.2402		-188248.2402
Receive										820000			3200000			
Expenditure	37	0	0	0	0	0	0	0	0		2089500	2364375		-1008248.24		-1008248.24
Receive										0			3200000			

Table 6: Cash Flow Analysis for Variation Scenario 2 – Financial Variation (Interest Rates)

Variation Scenario II	Week	Direct Cost	Indirect Cost	Insurance	Risk Allocation	Subtotal - not included risk - not paid by the owner in this scenario	Markup (25%)	Total Billed	Retainage Applied (100%)	Payment Received	Total Cost to Date - Including Risk	Total Billed to Date	Total Paid to Date	Overdraft End of Month	Interest (20%)	Max Overdraft (to date)
Expenditure	1	19000	14375	19000	5500	52375	13093.75	65468.75	65468.75	500000	57875	65468.75	500000	-442125	0	-442125
Receive																
Expenditure	2	19000	14375	0	5500	33375	8343.75	41718.75	41718.75		96750	107187.5		-403250	0	-403250
Receive										0			500000			
Expenditure	3	28000	14375	0	5500	42375	10593.75	52968.75	52968.75		144625	160156.25		-355375	0	-355375
Receive										0			500000			
Expenditure	4	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		208500	233125		-291500	0	-291500
Receive										0			500000			
Expenditure	5	34000	14375	0	5500	48375	12093.75	60468.75	60468.75		262375	293593.75		-237625	0	-237625
Receive										0			500000			
Expenditure	6	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		318250	356562.5		-181750	0	-181750
Receive										0			500000			
Expenditure	7	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		374125	419531.25		-125875	0	-125875
Receive										300000			800000			
Expenditure	8	28000	14375	0	5500	42375	10593.75	52968.75	52968.75		422000	472500		-378000	0	-378000
Receive										0			800000			
Expenditure	9	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		485875	545468.75		-314125	0	-314125
Receive										0			800000			
Expenditure	10	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		558750	629687.5		-241250	0	-241250
Receive										0			800000			
Expenditure	11	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		631625	713906.25		-168375	0	-168375
Receive										0			800000			
Expenditure	12	36000	14375	0	5500	50375	12593.75	62968.75	62968.75		687500	776875		-112500	0	-112500
Receive										0			800000			
Expenditure	13	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		768375	871093.75		-31625	0	-31625
Receive										0			800000			
Expenditure	14	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		849250	965312.5		49250	9850	59100
Receive										340000			1140000			
Expenditure	15	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		913125	1038281.25		-217025	0	-217025
Receive										0			1140000			
Expenditure	16	53000	14375	0	5500	67375	16843.75	84218.75	84218.75		986000	1122500		-144150	0	-144150
Receive										0			1140000			
Expenditure	17	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1066875	1216718.75		-63275	0	-63275
Receive										0			1140000			
Expenditure	18	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1138750	1299687.5		8600	1720	10320
Receive										0			1140000			
Expenditure	19	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1210625	1382656.25		82195	16439	98634
Receive										0			1140000			
Expenditure	20	51000	14375	0	5500	65375	16343.75	81718.75	81718.75		1281500	1464375		169509	33901.8	203410.8
Receive										380000			1520000			
Expenditure	21	51000	14375	0	5500	65375	16343.75	81718.75	81718.75		1352375	1546093.75		-105714.2	0	-105714.2
Receive										0			1520000			
Expenditure	22	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1433250	1640312.5		-24839.2	0	-24839.2
Receive										320000			1840000			
Expenditure	23	61000	14375	0	5500	75375	18843.75	94218.75	94218.75		1514125	1734531.25		-263964.2	0	-263964.2
Receive										0			1840000			
Expenditure	24	52000	14375	0	5500	66375	16593.75	82968.75	82968.75		1586000	1817500		-192089.2	0	-192089.2
Receive										350000			2190000			
Expenditure	25	44000	14375	0	5500	58375	14593.75	72968.75	72968.75		1649875	1890468.75		-478214.2	0	-478214.2
Receive										0			2190000			
Expenditure	26	34000	14375	0	5500	48375	12093.75	60468.75	60468.75		1703750	1950937.5		-424339.2	0	-424339.2
Receive										0			2190000			
Expenditure	27	32000	14375	0	5500	46375	11593.75	57968.75	57968.75		1755625	2008906.25		-372464.2	0	-372464.2
Receive										0			2190000			
Expenditure	28	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1808500	2068125		-319589.2	0	-319589.2
Receive										0			2190000			
Expenditure	29	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1861375	2127343.75		-266714.2	0	-266714.2
Receive										360000			2550000			
Expenditure	30	33000	14375	0	5500	47375	11843.75	59218.75	59218.75		1914250	2186562.5		-573839.2	0	-573839.2
Receive										0			2550000			
Expenditure	31	16000	14375	0	5500	30375	7593.75	37968.75	37968.75		1950125	2224531.25		-537964.2	0	-537964.2
Receive										330000			2880000			
Expenditure	32	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		1978000	2252500		-840089.2	0	-840089.2
Receive										0			2880000			
Expenditure	33	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2005875	2280468.75		-812214.2	0	-812214.2
Receive										0			2880000			
Expenditure	34	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2033750	2308437.5		-784339.2	0	-784339.2
Receive										0			2880000			
Expenditure	35	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2061625	2336406.25		-756464.2	0	-756464.2
Receive										0			2880000			
Expenditure	36	8000	14375	0	5500	22375	5593.75	27968.75	27968.75		2089500	2364375		-728589.2	0	-728589.2
Receive										320000			3200000			
Expenditure	37	0	0	0	0	0	0	0	0		2089500	2364375		-1048589.2	0	-1048589.2
Receive										0			3200000			

5. Observations, Understanding and Discussion

I have not made any changes to the overall project plan because, based on the results from the Monte Carlo simulations and the cash flow analysis under the Original Scenario, the project is still completed within the planned 36 week duration established in Segment 2 (Project Scheduling and Sub-Contractor Selection). The cash flow results also show a positive overdraft trend, indicating that the project remains financially healthy and profitable. Additionally, the contingency amount was selected using an 85% confidence level, which provides an effective balance between risk preparedness and cost efficiency. Choosing a lower confidence level (such as 25%) could underestimate the financial impact of uncertainties and increase the likelihood of cost overruns. In contrast, a much higher confidence level (such as 95%) would require a larger contingency reserve, potentially tying up more funds than necessary. Therefore, 85% is a reasonable and widely accepted choice for a project of this size and complexity, ensuring preparedness for probable risks without placing unnecessary strain on the project budget.

Importance of Cash Flow Analysis: Cash flow analysis is a critical component of project financial management. It allows project managers to track when cash is spent and when revenue is received throughout the project timeline. Even if a project is profitable overall, poor cash flow timing can lead to negative cash balances, financing challenges, and delays in procurement or execution. By evaluating cash inflows and outflows over the project duration, we can determine how much working capital is required, identify periods of financial stress, and proactively plan to avoid liquidity issues. This ensures that the project can progress smoothly without interruptions caused by funding shortages.

In addition to the Original Scenario, I assessed two variation scenarios to understand how changes in payment structure and financing conditions affect the project's cash flow:

- **Variation Scenario 1 (Payment Variation – No Mobilization Payment):**
In this scenario, the project begins without the initial mobilization payment. As a result, the cash balance remains negative until approximately Week 31, since expenses are incurred continuously while payments are only received after completion of each project. This scenario significantly increases financial strain and shows a noticeably deeper and prolonged overdraft curve. Therefore, it is not considered optimal from a project financing standpoint.
- **Variation Scenario 2 (Financial Variation – Interest Rate Increased to 20%):**
In this scenario, the mobilization payment is retained, but the interest rate is increased from 1.5% to 20%. While this does increase financing costs, the impact on the overall overdraft curve is not as severe as in Variation Scenario 1. The change is present but not dramatically visible unless examined closely in the financial details.

Out of curiosity, I also tested an additional combination of no mobilization payment and 20% interest, which resulted in a highly unfavorable financial situation; however, this case is not included in the report as it was explored only for comparative understanding.